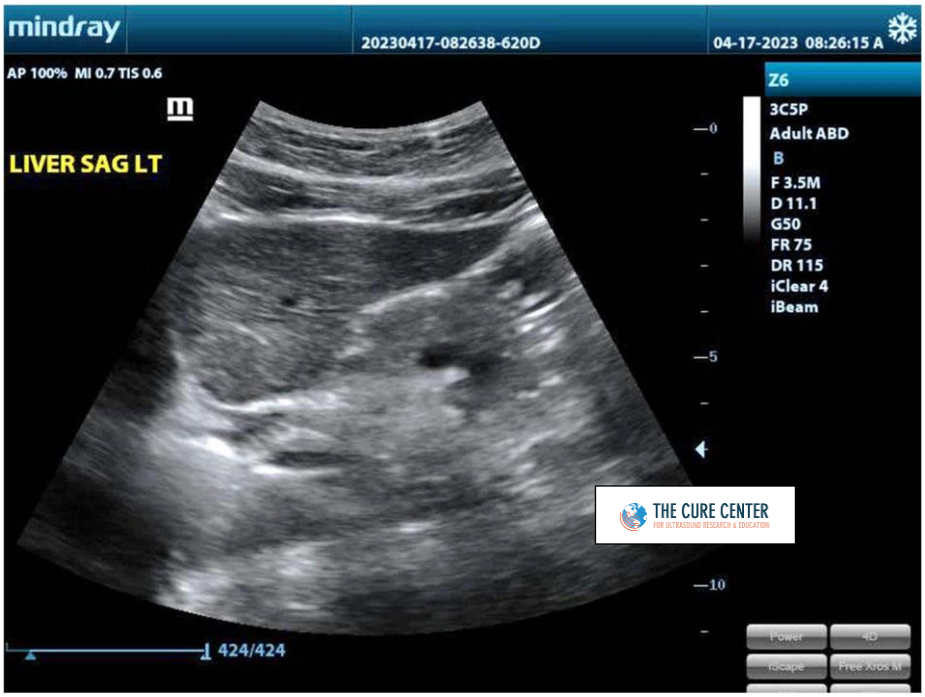
Liver Protocol

LIVER SAG LT:

1. SAG Left Lobe – Tip (the inferior margin)

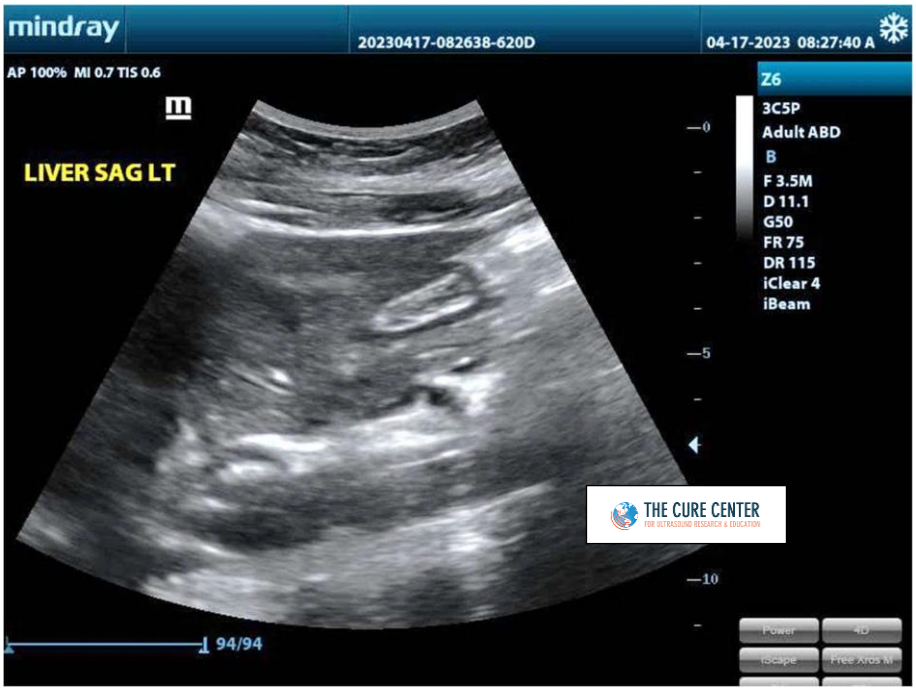
Note: your annotations should match the annotations in **blue (title above)**.

Note: Put the transducer in the middle of the stomach and angle out to the left of the patient's body.



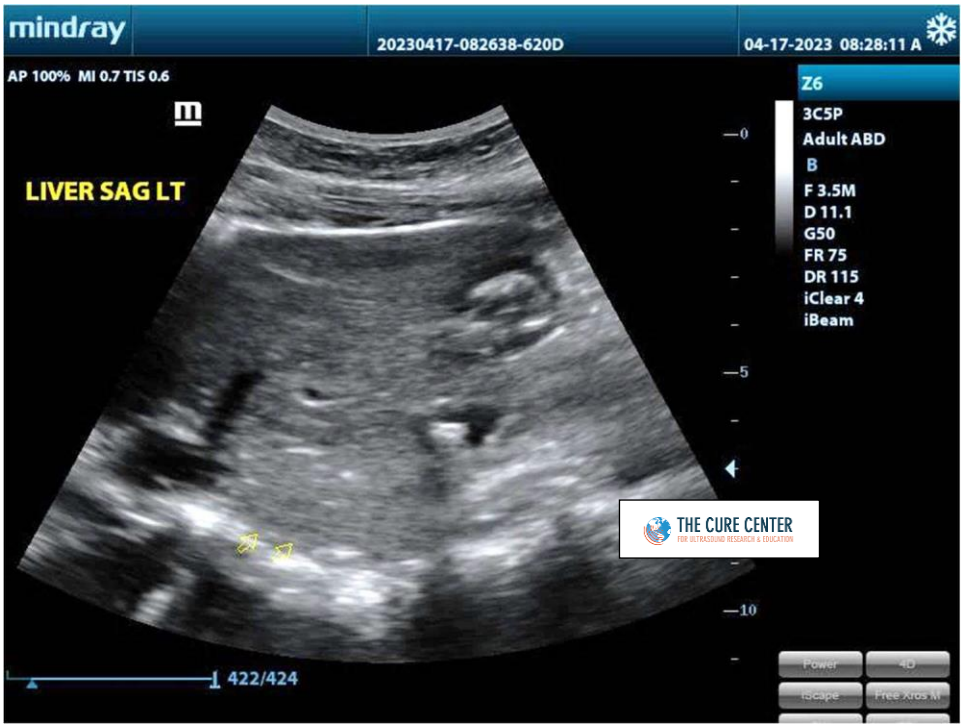
2. SAG Left Lobe – Aorta

Note: Stay where you are and start scanning a little to the patient's right



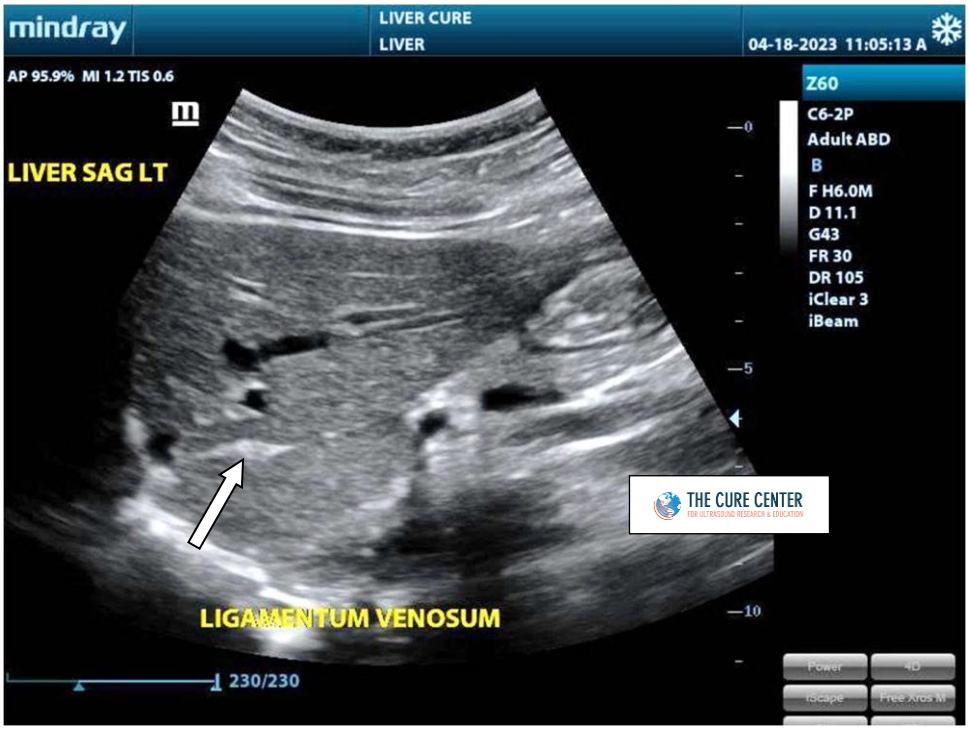
3. SAG Left Lobe – Caudate Lobe

Note: Stay where you are and scan more to the patient's right. Pass the IVC and decrease the depth



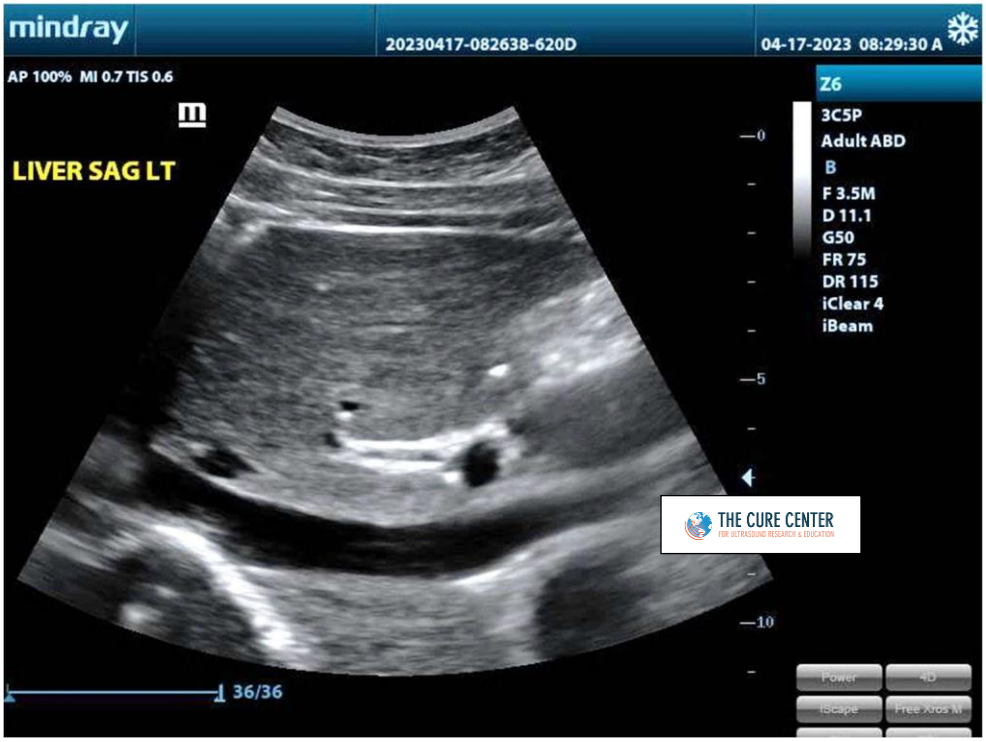
4. SAG Left Love – Ligamentum Venosum

Note: Pass the IVC and decrease the depth



5. SAG Left Lobe – Inferior Vena Cava (IVC)

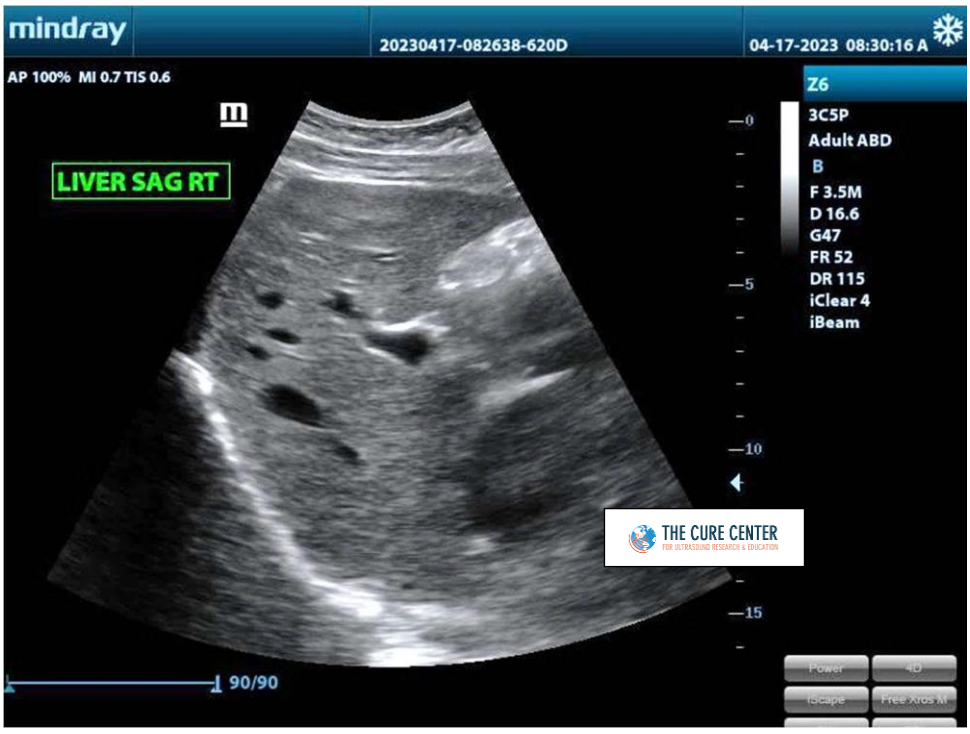
Note: Stay where you are and come even more to the patient's right until you see the heart moving. Decrease the depth



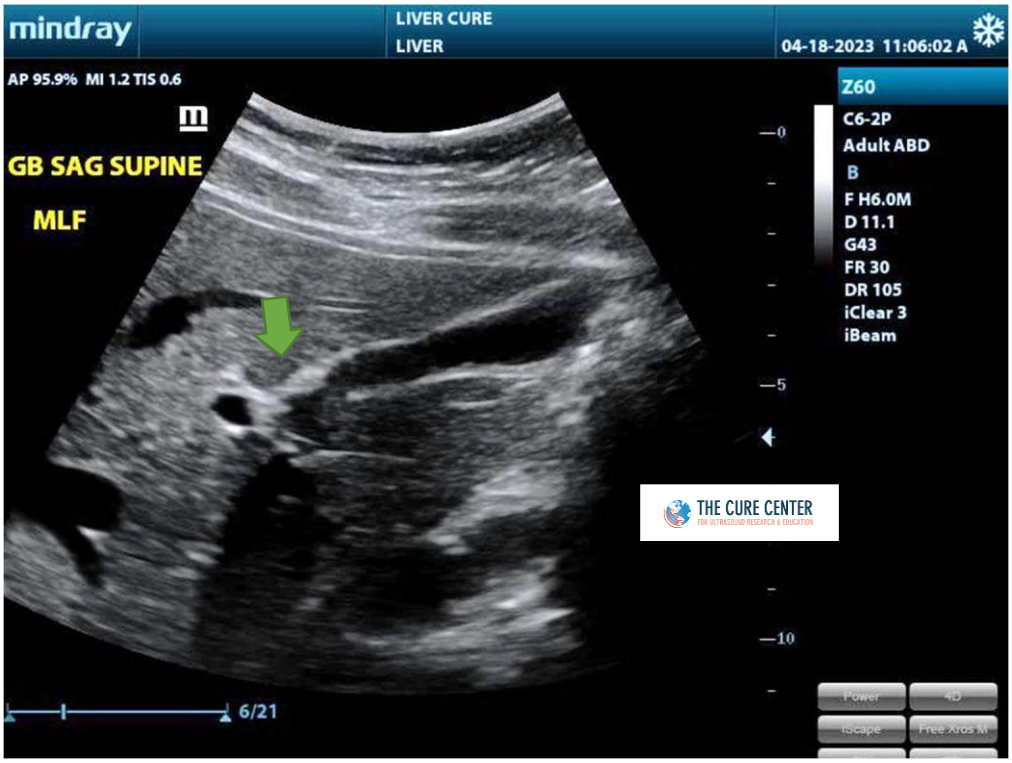
LIVER SAG RT:

1. SAG Right Lobe - Long Liver

Note: Decrease the depth while you are scanning to the patient's right. You will obtain the left and right liver with some dome.



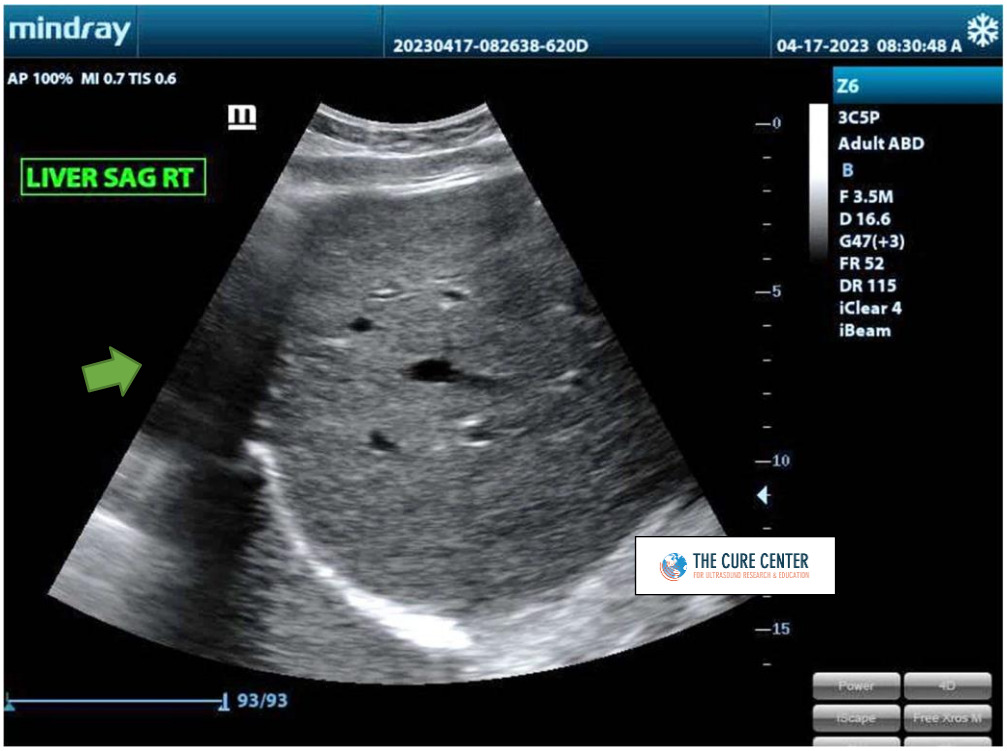
2. SAG Right Lobe – Main Lobar Fissure (MLF)



Note: Decrease the depth. Find the GB in the sag and connect it to the PV. The PV should look round.

FOR THE NEXT IMAGES 3,4, AND 5, ANGLE THE PROBE PERPENDICULAR TO THE DIRECTION OF THE RIGHT RIBS. SLIDE DOWN THE RIBS TO FIND 3, 4, AND 5.

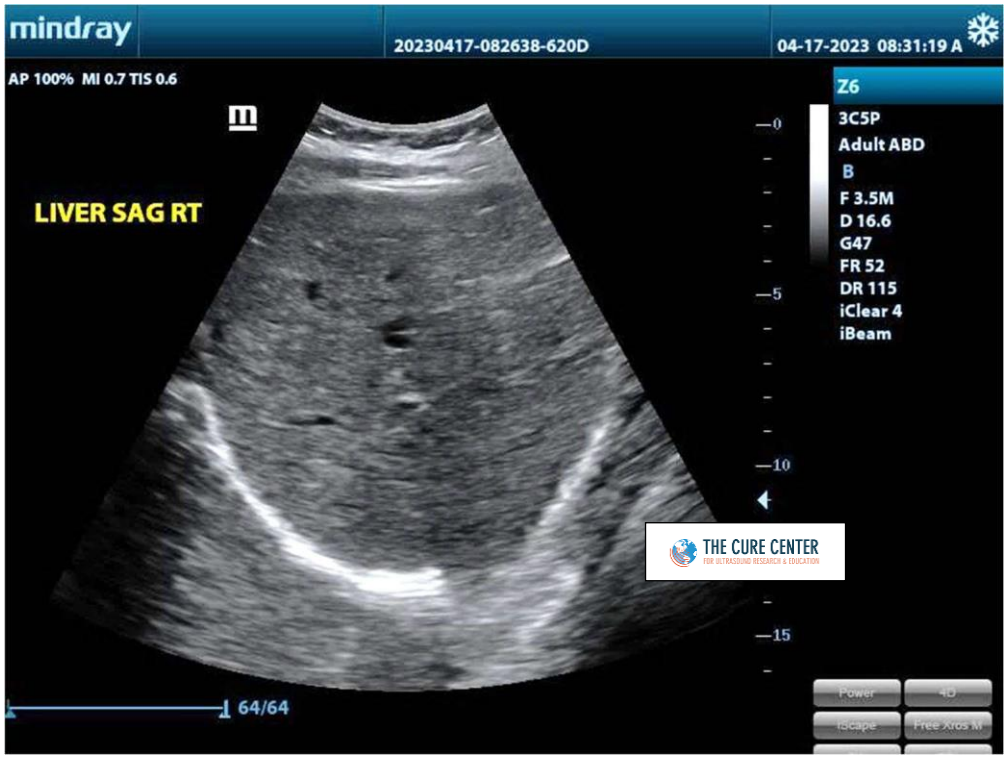
3. SAG Right Lobe – Dome w/ Heart



Note: Turn the patient away from you LLD for better images. Make sure the GB is not in the liver image. Close the dome of the liver to see the heart.

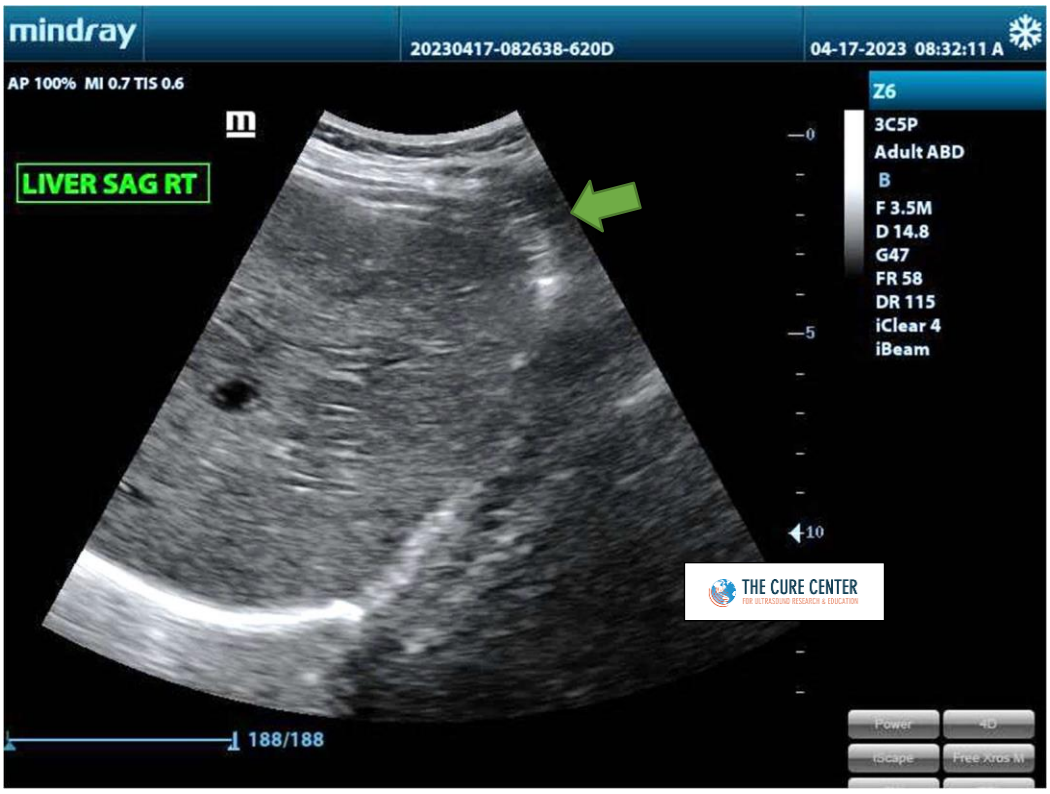
4. SAG Right Lobe – Dome Centered

Note: Turn the patient to the his/her left (LLD) for better images. Make sure the GB is not in the liver image. Center the liver to see a complete view of the dome.



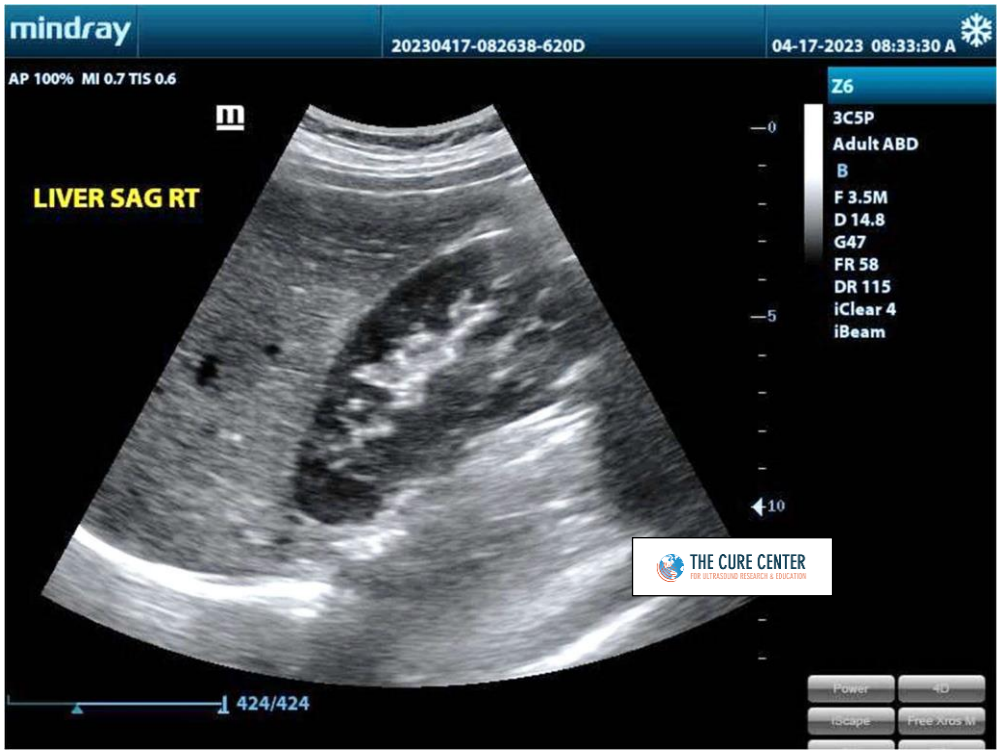
5. SAG Right Lobe – Dome w/ Corner

Note: Turn the patient to his/her left (LLD) for better images. Make sure the GB is not in the liver image.
Slide down more inferior to see the other corner of the liver.



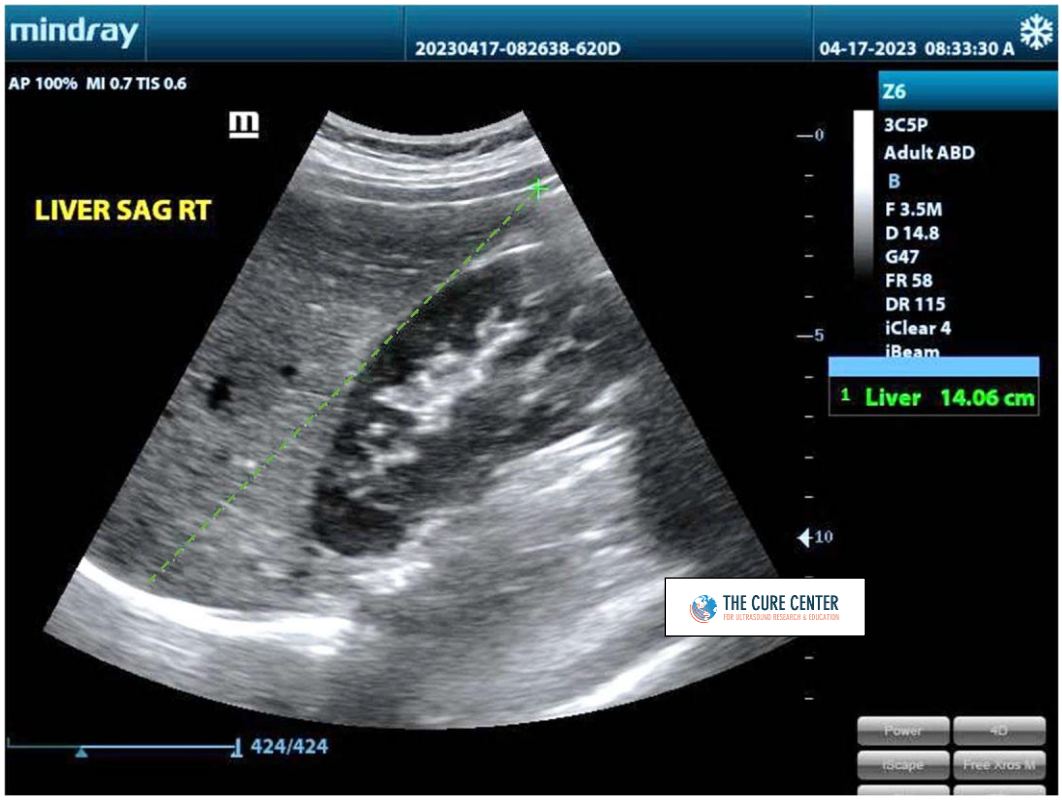
6. SAG Right Lobe – Liver/ Kidney Interface

Note: Place the transducer inferior to the right ribs, then turn the notch toward the patient's head. Make sure you see the tip of the liver to measure the liver.



7. SAG Right Lobe – Liver/ Kidney Interface w/ Measurement

Note: Stay where you are and measure the liver from the tip to the dome inner to inner.



LIVER TRV LT:

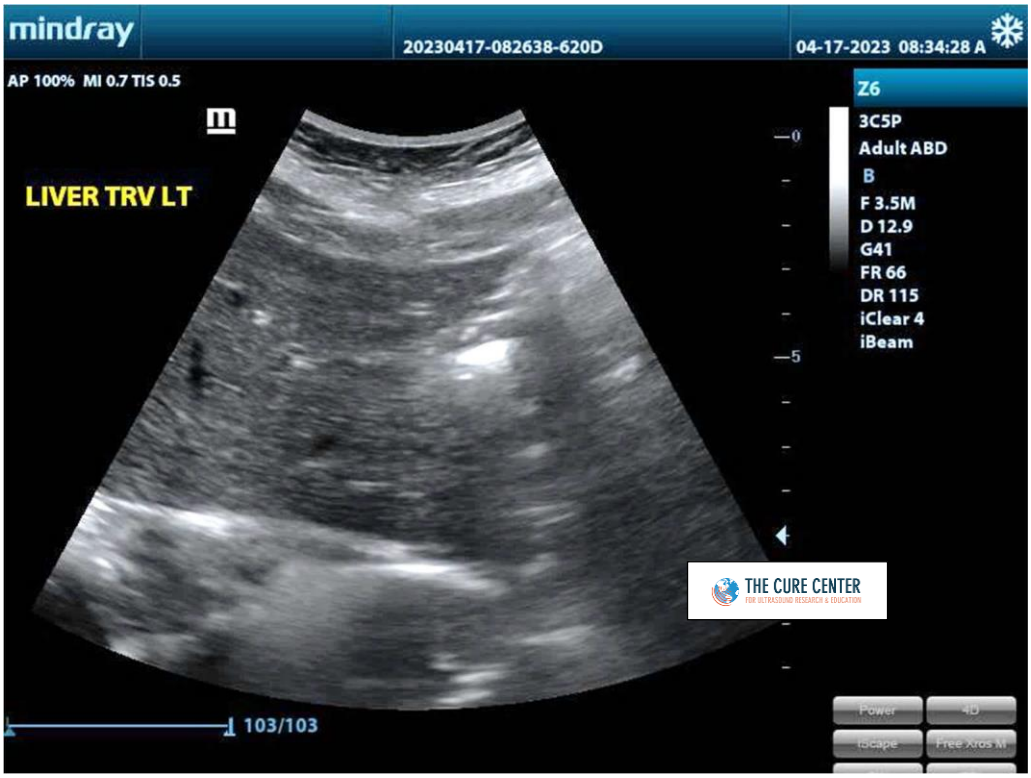
1. TRV Left Lobe – Ligamentum Teres

Note: Place the transducer toward the left chest. Keep going up and down on the left lobe to find the echogenic round spot.



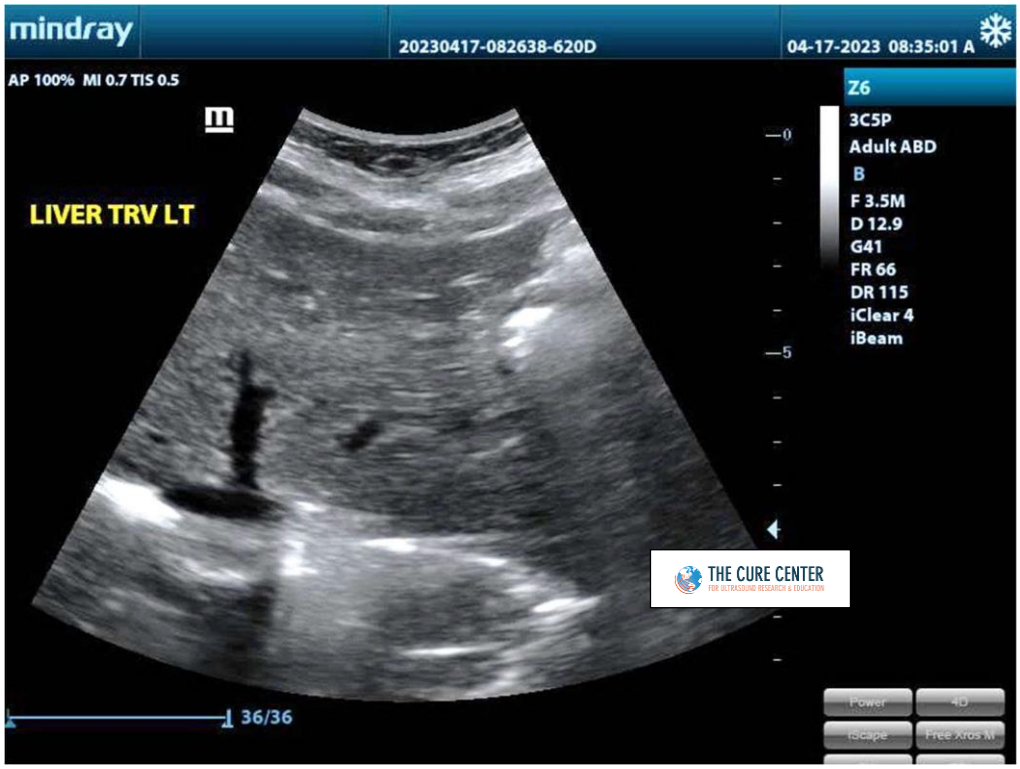
2. TRV Left Lobe - Tip (the inferior margin-toward the patient's feet)

Note: Stay where you are and angle transducer superiorly towards the left chest. The liver Left Tip should look triangular.



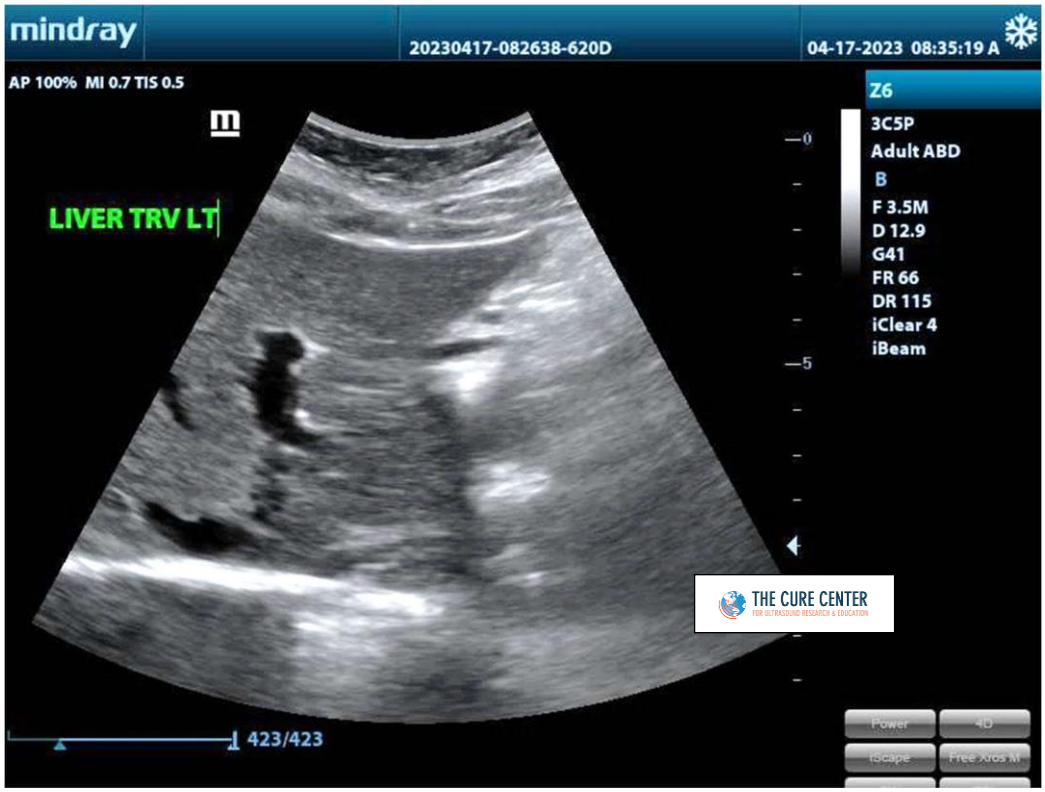
3. TRV Left Lobe – Left Hepatic Vein (LHV)

Note: Move the transducer a little down to find the left hepatic vein coming off the IVC.



4. TRV Left Lobe – Left Portal Vein (LPV)

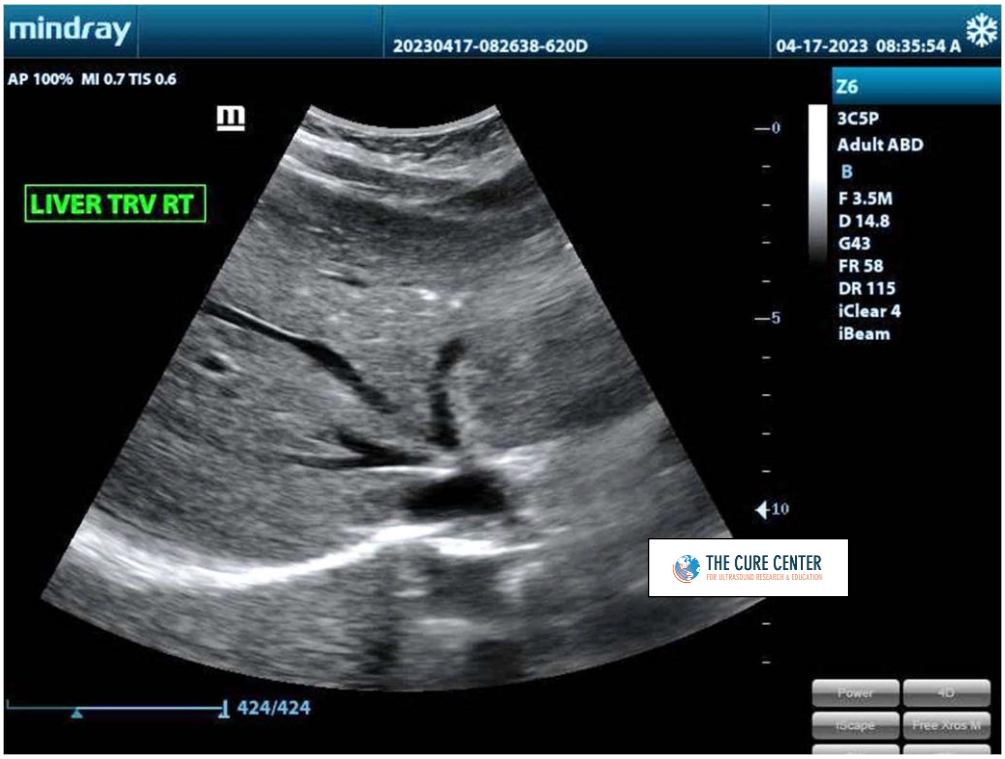
Note: Move the transducer further down to find the left portal vein.



LIVER TRV RT:

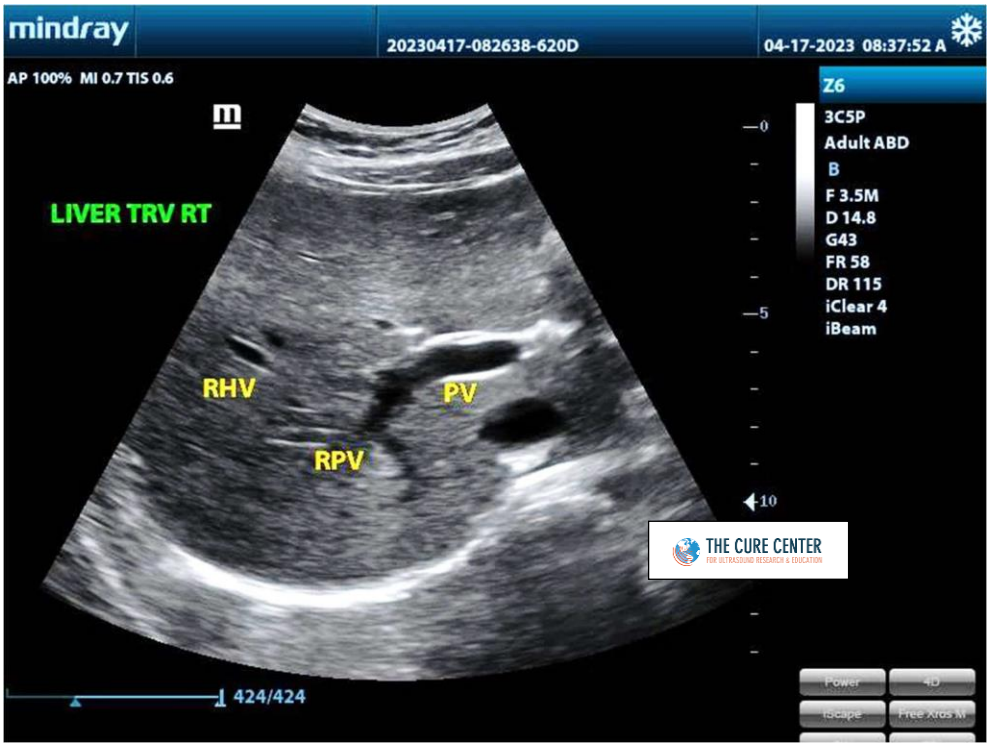
1. TRV Right Lobe – Hepatic Veins/ Rt, Mid, Lt

Note: angle the transducer towards the right rib (notch is facing the right) to find the hepatic veins. They all have to be coming off the IVC (sometimes you will see only two HV out of three).



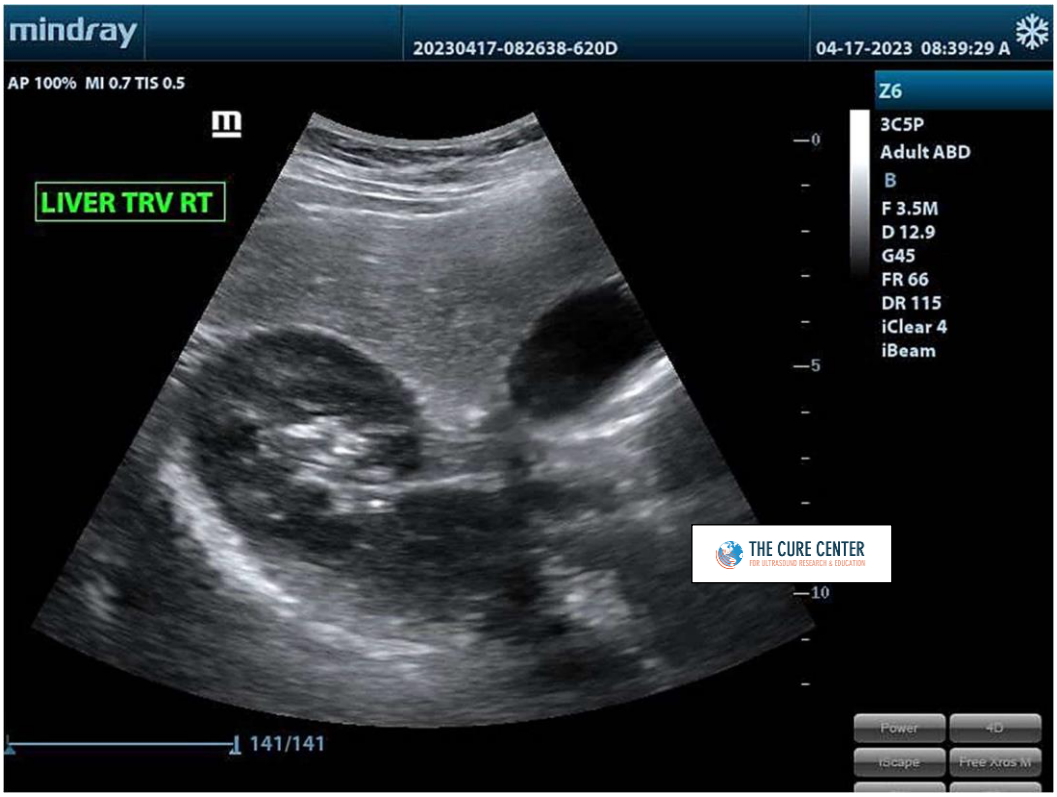
2. TRV Right Lobe – RHV/ RPV/ PV

Note: Place the transducer's notch toward you. The probe needs to be perpendicular to the right ribs.



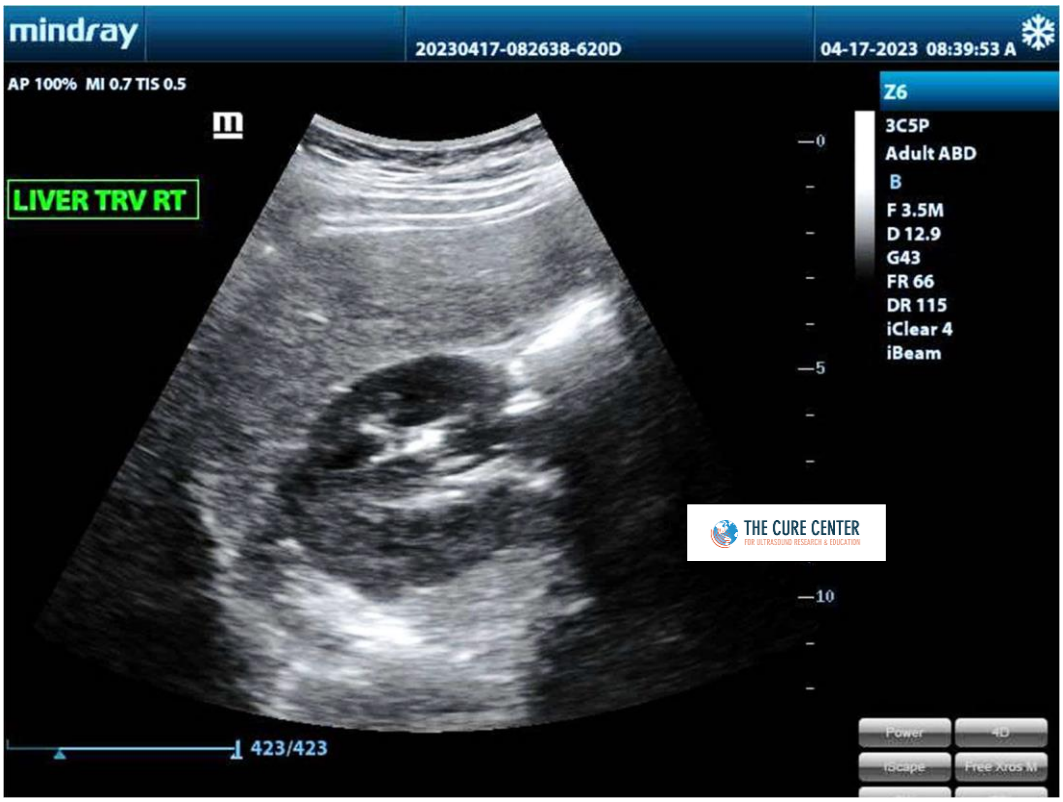
3. TRV Right Lobe – Liver/ Kidney/GB

Note: Stay where you are and slide the prove further down inferior to see the GB, when you find it, twist your transducer to get the right kidney in the image at the same time.



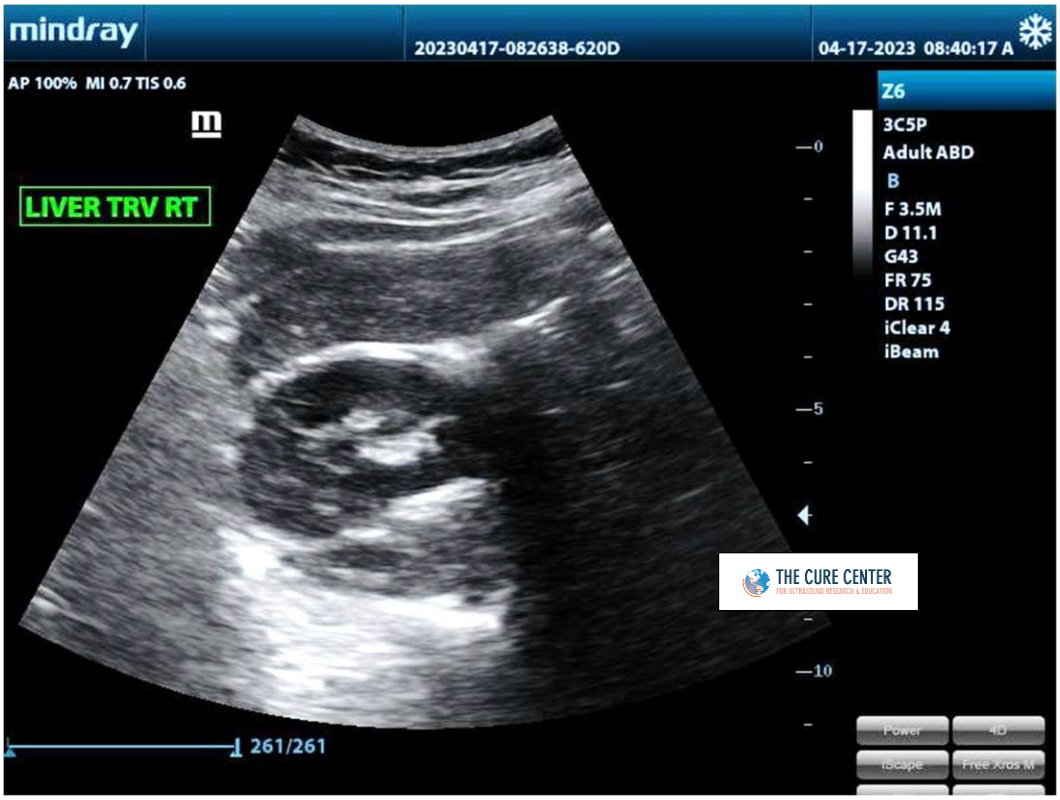
4. TRV Right Lobe – Liver/Kid

Note: Slide down more inferiorly and remove the GB from your picture to get this image.



5. TRV Right Lobe – Liver Small Piece

Note: Slide more down and inferiorly to make your liver look smaller.



6. TRV Right Lobe – PV No Color

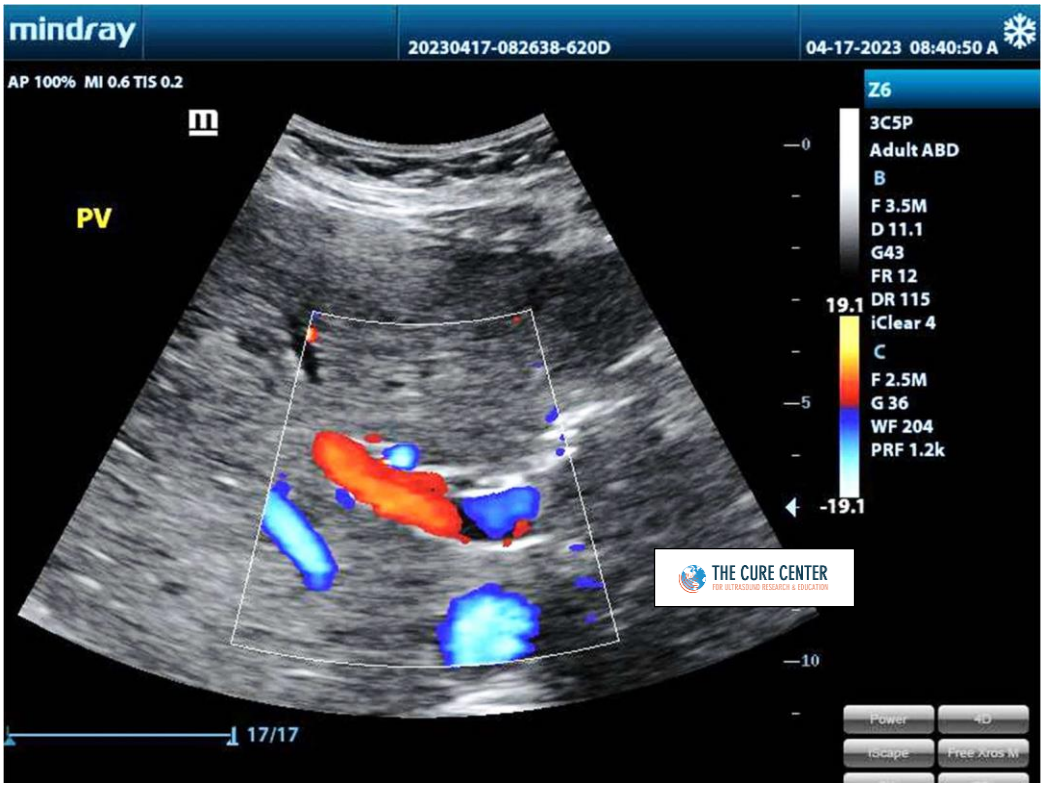
Note: Your transducer will be transverse while the portal vein is elongated.

Note: Make sure that half of the prove is on the right ribs and other half on the patient's belly. PV needs to look red, also on the US the doppler spectral waveform is above baseline.



7. TRV Right Lobe – PV w/ Color

Note: Make sure that half of the prove is on the right ribs and other half on the patient's belly. PV needs to look red, also on the US the doppler spectral waveform is above baseline.



8. TRV Right Lobe – PV Color Doppler

Note: Your transducer will be transverse while the portal vein is elongated. The Doppler needs to be straight down. **No angle is needed** with the portal vein.

Note: Make sure that half of the prove is on the right ribs and other half on the patient's belly. PV needs to look red, also on the US the doppler spectral waveform is above baseline.

